SEÁN Ó MURCHÚ

Senior Software Engineer

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Backend engineer with eight years of experience designing distributed, event-driven systems. Specialises in functional programming, observability, and developer experience.

EXPERIENCE

Senior Software Engineer

Acme Corp

Jun 2022 – Present Dublin, Ireland

Platform team lead. Owns the event-sourcing stack underpinning the wider product surface.

- Led the migration of a customer-facing monolith into a set of event-driven microservices, halving p99 latency.
- Designed and rolled out an event-sourcing platform now used by four product teams.
- Mentored three engineers from mid to senior level.

Software Engineer

Liffey Labs

Jun 2019 – Jun 2021 Remote

- Built and shipped the first version of a SaaS product from scratch alongside a two-person team.
- Introduced the CI/CD pipeline and developer-onboarding flow that scaled the engineering org from 3 to 15.

VOLUNTEER

Mentor

CoderDojo Dublin

Jun 2020 – Present

Volunteer-led coding club introducing young people to programming.

- Weekly mentoring sessions for 10–14 year-olds learning to code.
- Curate the Scala / functional-programming track.

PROJECTS

Application

Hyperion

Acme Corp

Distributed task scheduler in Rust

Lead Architect

Jan 2024 – Aug 2024

- Handled 10k tasks/s on a single node.
- Designed an idempotent retry protocol.

Rust Tokio gRPC

AREAS OF FOCUS

- Distributed systems: Kafka, CDC, idempotent consumers, event sourcing.
- Functional programming and type-driven design in Scala and Haskell.
- Developer experience: tooling, CI/CD, observability.
- Engineering culture: ADRs, RFCs, mentoring.

SKILLS

Languages - Scala Haskell Python Go Rust

Infra - Kafka AWS Terraform Docker Kubernetes

LANGUAGES

English ●●●●●○

Irish ●●●○○○

French ●●●●○○

Portuguese ●●◐○○○

EDUCATION

M.Sc. in Computer Science

Tallaght Institute of Technology

2017 – 2019

Distinction Distributed Systems Type Theory Concurrency

Computer Science

Trinity College Dublin

2014 – 2017

Crucible

Migration tool for legacy databases

Maintainer

Sep 2024 – Present

- Schema diffing across PostgreSQL and MySQL.

Library

Tinkerbell

Tiny IRC bot

Lua

Other

Quill

PUBLICATIONS

Articles

- *Event Sourcing in Practice*.
  Personal Blog
  14 Jun 2024
  A walk-through of idempotent Kafka consumers and the replay strategy that ships with them.
- *Untyped Note*.
  Jan 2024

Conference Papers

- *Idempotent Kafka Consumers at Scale*.
  EuroSys
  Apr 2024

Talks

- *Designing for Failure*.
  ScalaIO
  Sep 2023

Books

- *Functional Domain Modelling*.
  Self-published
  2022

INTERESTS

Music - Trad Jazz

Sport - Hurling Climbing

Travel - Japan Iceland

CERTIFICATIONS

CNCF

Certified Kubernetes Administrator · Aug 2023

Certified Kubernetes Application Developer · Jan 2024

AWS

AWS Solutions Architect — Professional · May 2023

AWS DevOps Engineer — Professional · Sep 2024

Hashicorp Terraform Associate · Dec 2022

Certified Kubernetes Security Specialist · Mar 2024

Google Cloud Professional Architect · Nov 2023

AWARDS

Best Paper Award

ScalaConf

Sep 2024

For *Idempotent Kafka Consumers*.

Dean's List

REFERENCES

Dr Ada Lovelace

Seán is a calm, methodical engineer who left every system he touched in a better state than he found it. Recommended without reservation.

Pragmatic, curious, unfailingly kind — would hire again in a heartbeat.